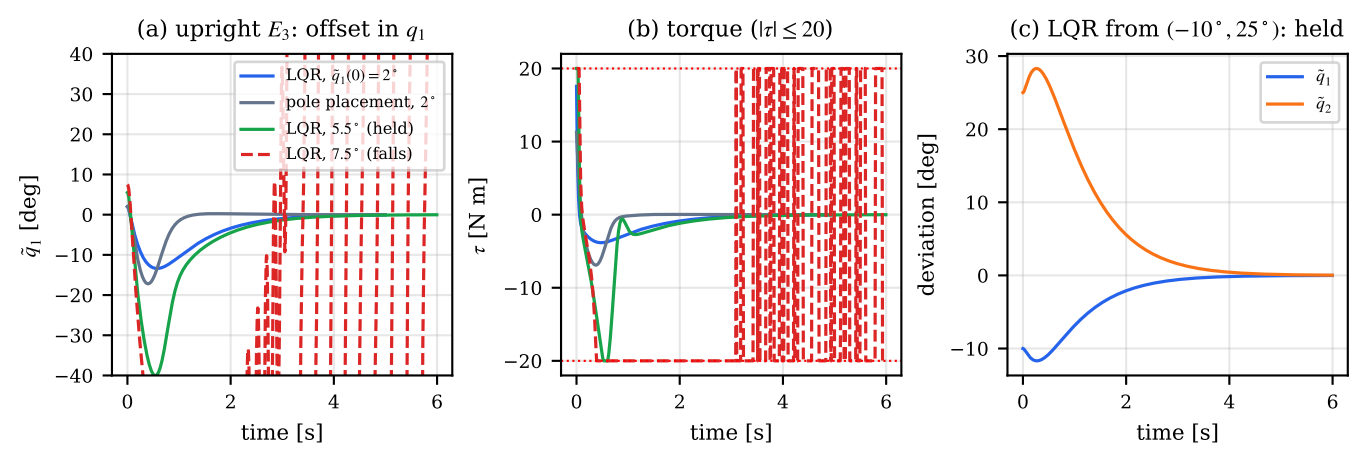
(a) upright E_3: offset in q_1
LQR, $\tilde{q}_1(0) = 2^\circ$
pole placement, 2°
LQR, 5.5° (held)
LQR, 7.5° (falls)
$\tilde{q}_1$ [deg]
time [s]
(b) torque ($|\tau| \le 20$)
τ [N m]
time [s]
(c) LQR from $(-10^\circ, 25^\circ)$: held
$\tilde{q}_1$
$\tilde{q}_2$
deviation [deg]
time [s]